

MONTREAL TRUDEAU, QC, Canada

WMO: 716270

Lat: 45.4667N

Lon: 73.7500W

Elev: 118

StdP: 14.63

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.1	-4.3	-21.1	1.7	-7.8	-16.0	2.3	-3.6	29.8	25.3	26.9	21.3	9.7	260	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	86.5	71.8	83.7	70.0	81.3	68.6	74.1	82.8	72.4	80.3	70.9	78.3	12.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.3	116.0	79.0	69.8	109.9	77.1	68.2	103.9	76.0	37.8	82.6	36.2	80.5	34.8	78.2	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.3	22.2	19.7	DB	-14.2	91.0	5.1	2.6	-17.9	92.9	-20.9	94.4	-23.8	95.9	-27.5	97.8	(4)
(5)			WB	-14.9	77.8	4.9	1.8	-18.4	79.1	-21.2	80.2	-24.0	81.2	-27.5	82.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	16.1	19.1	29.2	43.5	57.3	66.6	71.3	69.5	62.0	49.0	36.5	24.1		
	DBStd	21.18	12.56	10.79	10.30	8.36	7.67	6.58	5.03	5.35	7.27	7.99	8.87	10.79		
	HDD50	4145	1051	865	646	228	19	0	0	0	3	118	411	804		
	HDD65	7662	1516	1285	1109	646	260	58	7	19	141	498	855	1268		
	CDD50	2496	0	0	2	32	246	498	659	604	362	87	6	0		
	CDD65	542	0	0	0	1	22	106	201	159	51	2	0	0		
	CDH74	3601	0	0	1	13	191	750	1366	964	308	8	0	0		
	CDH80	810	0	0	0	1	36	180	330	200	63	0	0	0		
	WSAvg	9.5	10.5	10.2	10.1	10.5	9.6	8.7	8.5	8.1	8.3	9.4	9.8	10.1		
	PrecAvg	38.9	3.0	2.6	2.8	3.1	2.9	3.3	3.5	3.7	3.5	3.1	3.7	3.4		
(15) Precipitation	PrecMax	49.1	5.9	7.0	4.5	7.0	5.5	5.7	7.3	6.8	9.1	7.2	7.7	6.6		
	PrecMin	32.5	0.9	0.4	0.7	0.6	0.6	0.9	1.2	0.9	0.9	0.7	1.3	1.0		
	PrecStd	4.3	1.3	1.3	1.0	1.4	1.2	1.2	1.5	1.5	1.7	1.4	1.4	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.5	45.0	60.1	75.6	84.2	88.9	90.1	88.5	86.5	74.8	62.1	51.3		
		MCWB	45.1	40.8	49.7	60.2	66.3	72.8	74.3	73.2	72.3	63.5	56.6	48.7		
	2%	DB	40.8	40.4	51.2	67.5	79.2	84.9	86.5	84.6	81.1	69.5	57.2	45.1		
		MCWB	38.2	37.1	43.7	53.3	63.2	69.9	72.3	70.7	68.9	61.2	52.8	42.5		
	5%	DB	37.4	37.5	46.3	61.9	74.9	81.5	83.6	82.0	77.4	64.8	52.9	41.2		
		MCWB	35.3	34.8	39.9	50.3	60.3	67.9	70.7	69.3	67.2	57.5	48.4	38.6		
	10%	DB	34.0	34.9	42.5	56.9	70.8	78.4	81.0	79.4	73.9	61.0	48.9	37.8		
		MCWB	32.0	32.5	37.6	47.4	58.7	66.2	69.0	68.2	65.4	55.2	44.8	35.4		
	0.4%	WB	45.1	42.1	53.1	62.2	69.7	75.4	77.2	75.5	73.9	66.6	58.0	49.0		
		MCDB	46.0	44.7	58.3	72.6	80.3	85.2	87.5	85.7	82.3	72.4	60.5	51.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.0	37.7	45.0	56.0	66.3	72.5	74.6	73.1	71.0	61.9	53.3	42.9		
		MCDB	40.4	40.0	50.3	64.1	75.1	81.3	83.0	81.3	78.2	67.6	56.8	45.0		
	5%	WB	35.4	35.1	40.9	51.9	63.4	70.3	72.7	71.5	68.8	58.6	48.9	38.8		
		MCDB	37.2	37.2	45.7	60.2	71.3	78.4	80.4	78.7	75.8	63.8	52.3	40.7		
	10%	WB	32.2	32.7	37.9	48.3	60.5	68.2	71.0	70.0	66.5	55.6	45.1	35.6		
		MCDB	34.1	34.9	41.8	55.8	67.7	75.9	78.4	76.7	73.0	60.1	48.5	37.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.7	14.2	14.3	16.6	17.7	16.6	15.8	16.1	16.5	13.9	12.2	11.6		
		MCDBR	15.9	15.0	18.5	25.3	23.3	20.1	18.1	18.1	19.5	19.2	18.0	13.5		
		MCWBR	15.6	13.9	13.4	15.7	12.2	10.3	9.3	8.9	10.6	12.6	14.9	12.9		
		MCDBR	15.5	14.5	17.7	22.8	19.3	17.3	15.9	15.8	17.4	17.3	17.2	13.4		
	5% WB	MCWBR	15.4	13.9	14.1	15.8	12.3	10.5	9.6	9.1	10.8	12.7	15.4	13.0		
(40) Clear-Sky Solar Irradiance	taub		0.261	0.289	0.332	0.365	0.390	0.446	0.431	0.416	0.375	0.338	0.308	0.292		
	taud		2.344	2.257	2.290	2.358	2.348	2.196	2.246	2.284	2.385	2.494	2.489	2.308		
	Ebn at Noon		272	282	284	284	279	262	265	264	267	264	253	242		
	Edh at Noon		24	32	36	37	38	45	42	39	33	25	21	23		
(44) All-Sky Solar Radiation	RadAvg		455	758	1155	1391	1627	1785	1800	1592	1258	729	443	319		
	RadStd		52	77	95	158	155	136	136	74	103	65	56	26		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page